THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM TWO JOINT EXAMINATION FOR GOVERNMENT SCHOOLS – JEGOS

PRE-MOCK EXAMINATION
041 BASIC MATHEMATICS
MARKING SCHEME 2024

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1. (a) i. The largest number is 76 976.....01 mark
The smallest number is 71 .....01 mark
Quotient $= \frac{74\,976}{71} = 1056$.....01 mark
Therefore the quotient of the largest number and the smallest number is 1056
ii. The numbers are: 6 015, 102, 217, 2 662, 15 025, 725, 2352, and 5 675.....01 mark
Sum $= 6\,015 + 102 + 217 + 2\,662 + 15\,025 + 725 + 2\,352 + 5\,675$.....01 mark
$= 32\,773$.....01 marks

(b) Convert each subject performance into percentage for common and easy comparison
Chemistry score $\frac{14}{25} \times 100\% = 56\ \%$.....2nd position.....$00\frac{1}{2}$ mark
Mathematics score $\frac{5}{8} \times 100\% = 62.5\ \%$.....5th position.....$00\frac{1}{2}$ mark
English language score $= \frac{1}{2} \times 100\% = 50\ \%$.....1st position.....$00\frac{1}{2}$ mark
Physics score $= \frac{3}{5} \times 100\% = 60\%$.....3rd position.....$00\frac{1}{2}$ mark
Biology score $= \frac{31}{50} \times 100\% = 62\%$.....4th position.....$00\frac{1}{2}$ mark
Since $50\ \% < 56\ \% < 60\ \% < 62\ \% < 62.5\%$
Therefore his subjects wise performance in ascending order is given by English language, Chemistry, Physics, Biology, and Mathematics, ...$01\frac{1}{2}$ marks

2. (a) Distance covered $= 5.2\ km + 1.8\ km = 7\ km$ .....01 mark
Distance walked in 5 days to and from $= 7 \times 5 \times 2$.....01 mark
$= 70\ km$
Therefore he walks 70 km in 5 days .....02 marks

(b) i. $876\,346 \approx 880\,000$.....03 marks

ii. $(48\ l \quad 900\ ml) \div 3 = 16l \quad 300ml$

Therefore each family obtained $16\ l\ \ 300\ ml$.....03 marks

3. (a) Distance $= 50\ m + 80\ m + 50\ m + 12\ m = 80\ m$

$= 272\ m$ .....01 mark

$1\ m = 100\ m$

$272\ m = y$

$y = \frac{272\ m \times 100\ cm}{1m}$

$y = 27\ 200\ cm$.....01 mark

Number of rolls $= \frac{27200\ cm}{650\ cm} = 41.846$ rolls .....01 mark

Therefore $41.846\ rolls \approx 42\ rolls$

Therefore 42 rolls of fencing tape are needed to fence the field completely.....02 marks

(b) $m = 180^{\circ} - (73^{\circ} - 31^{\circ})$.....02 marks

$= 180^{\circ} - 42^{\circ}$.....01 mark.

$= 138^{\circ}$.....02 marks

4. (a) i. $Area(A) = length(l) \times width(w)$.....$00\frac{1}{2}$ mark

$A = l \times w$

$81a^2 - 1 = [9a]^2 - [1]^2$.....$00\frac{1}{2}$ mark

$= [9a + 1][9a - 1]$.....01 mark

Therefore the possible length of a rectangle is $[9a + 1]\ units$ and the width is $[9a - 1]\ units$ .....01 mark.

ii. $[2x - 5][x - 3] = 0\ is\ the\ same\ as\ 2x^2 - 11x + 15 = 0$

Therefore the equation is $2x^2 - 11x + 15 = 0$.....01 mark

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{--11 \pm \sqrt{11^2 - 4 \times 2 \times 15}}{2 \times 2}$

$= \frac{11 \pm \sqrt{121 - 120}}{4}$.....01 mark

$= \frac{11 \pm \sqrt{1}}{4}$

$= \frac{11 \pm 1}{4}$.....01 mark

$= \frac{11+1}{4}\ or\ \frac{11-1}{4}$

$= \frac{12}{4}\ or\ \frac{10}{4}$

$x = 3\ or\ x = 2.5$.....01 mark

(b) $\begin{cases} 5x - 2y = 12 \\ 2x + 3y = 1 \end{cases}$

$3 \mid 5x - 2y = 12$

$2 \mid 2x + 3y = 1$

$15x - 6y = 36$

$4x + 6y = 2$

$+\left|\begin{matrix}15x-6y=36\\4x+6y=2\end{matrix}\right.$

$19x=38$ .....01 mark

$\frac{19x}{19}=\frac{38}{19}, x=2$.....01 mark

Substitute 2 for $x\ in\ 2x+3y=1$

$2\times 2+3y=1,\ 3y=1-4, 3y=-3$

$\frac{3y}{3}=\frac{-3}{3}, y=-1$.....01 mark

5. (a) Percentage profit $=\left[\frac{Selling\ price-buying\ price}{buying\ price}\right]100\%$.....$00\frac{1}{2}$ mark

Let y be the buying price or cost price

$20\%=\left[\frac{60\ 000-y}{y}\right]100\%$.....$00\frac{1}{2}$ mark.

$\frac{20\%}{100\%}=\frac{60\ 000-y}{y}\times\frac{100\%}{100\%}$

$\frac{1}{5}=\frac{60\ 000-y}{y}$.....01 mark

$y=300\ 000-5y$

$y+5y=300\ 000$

$6y=300\ 00$.....01 mark

$\frac{6y}{6}=\frac{300\ 000}{6}$

$y=50\ 000$ shillings

Therefore the cost price is 50 000 shillings.....01 mark

(b) Total parts $=9+16+11=36\ parts$.....$00\frac{1}{2}$ mark

The sum of exterior angles of any polygon $=360^{\circ}$.....$00\frac{1}{2}$ mark

First exterior angle $=\frac{9}{36}\times 360^{\circ}=90^{\circ}$.....$00\frac{1}{2}$ mark

Second exterior angle $=\frac{16}{36}\times 360^{\circ}=160^{\circ}$.....$00\frac{1}{2}$ mark

Third exterior angle $=\frac{11}{36}\times 360^{\circ}=110^{\circ}$.....$00\frac{1}{2}$ mark

Interior angle + Exterior angle $=180^{\circ}$.....$00\frac{1}{2}$ mark

First interior angle $=180^{\circ}-90^{\circ}=90^{\circ}$.....$00\frac{1}{2}$ mark

Second interior angle $=180^{\circ}-160^{\circ}=20^{\circ}$.....$00\frac{1}{2}$ mark

Third interior angle $=180^{\circ}-110^{\circ}=70^{\circ}$.....$00\frac{1}{2}$ mark

Therefore the triangle a right angled triangle since one of its interior angles is 90°...$00\frac{1}{2}$ mark

6. (a) $m=\frac{y_2-y_1}{x_2-x_1}$.....01 mark

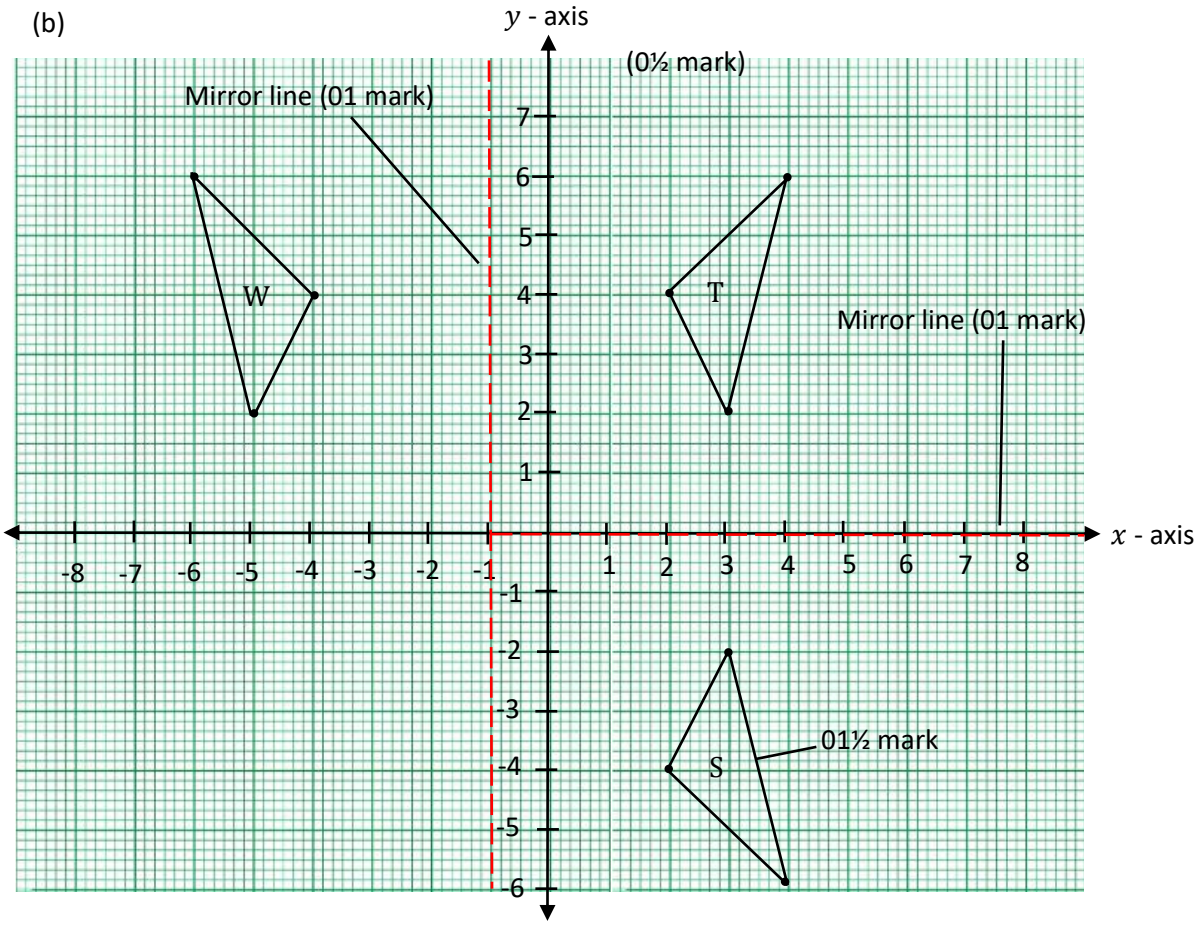
$\frac{1}{2}=\frac{2+2}{5-x}$.....01 mark

$2\times 4=5-x$.....01 mark.

$x=5-8$

$=-3$.....02 mark

(b)



7. (a) i. $0.000000212 = 2.12 \times 10^{-7}$.....02 marks

ii. $\log_2 8\left[\sqrt{\frac{a^5 b}{c^3}}\right] = \log_2 2^3 + \frac{1}{2}[\log_2 a^5 + \log_2 b - \log_2 c^3]$.....$00\frac{1}{2}$ mark

$= 3\log_2 2 + \frac{1}{2}[5\log_2 a + \log_2 b - 3\log_2 c]$.....01 mark

$= 3 + \frac{1}{2}[5 \times 3 - 2 - 3 \times 5]$...

$= 3 + \frac{1}{2}[15 - 2 - 15]$.....$00\frac{1}{2}$ mark

$= 3 + \frac{1}{2} \times -2$.....01 mark

$= 3 - 1$

$= 2$.....01 mark

(b) $2^x \div 2^{-2y} = 2^5$

$\frac{2^x}{2^{-2y}} = 2^5$

$x + 2y = 5$ ....$00\frac{1}{2}$ mark

$2^{2x} \times 2^{2y} = 2^4$

$2x + 2y = 4$

$x + y = 2$ ……..equation 2……$00\frac{1}{2}$ mark

Solve equation 1 and 2 by elimination method

$\begin{cases} x + 2y = 5 \\ x + y = 2 \end{cases}$

$-\begin{vmatrix} x + 2y = 5 \\ x + y = 2 \end{vmatrix}$

$y = 3$……01 mark

Substitute 3 for$y\ in\ \ equation\ 2$

$x + y = 2$

$x + 3 = 2$

$x = 2 - 3 = -1$ ……01 mark

8. (a)The Side$-\ Side - Side\ (SSS)$ theorem used in proving congruence of polygons implies equality of corresponding sides while the $Side - Side - Side\ (SSS)$ theorem used in proving similarity of polygons means the ratio of corresponding edges or sides is constant or proportional……03 marks

(b) (i) $\frac{AB}{AM} = \frac{BC}{MN} = \frac{AC}{AM} = Constant\ of\ proportionality$ ……$00\frac{1}{2}\ mark$

$\frac{BC}{MN} = \frac{AB}{AM}$ ……$00\frac{1}{2}\ mark$

$\frac{4.5}{MN} = \frac{6}{4}$

$\frac{4.5}{MN} = \frac{3}{2}$ ……$00\frac{1}{2}\ mark$

$2 \times 4.5 = 3\ MN$

$9 = 3\ MN$

$\frac{9}{3} = \frac{3MN}{3}$

$\overline{MN} = 3\ cm$ ……$00\frac{1}{2}\ mark$

(ii) $< BAC =< MAN\ (Common\ angle)$ …… 01 mark

$< AMN \cong < ABC\ (Corresponding\ angles\ given\ that(\overline{MN}\ \|\overline{BC}\ ))$ ……$00\frac{1}{2}\ mark$

$< AMN\ \cong < ABC\ (Corresponding\ angles\ given\ that(\overline{MN}\ \|\overline{BC}\ 0)$ ……$00\frac{1}{2}\ mark.$

$\therefore \Delta ABC \sim \Delta AMN\ (AA - Similarity\ theorem)$ ……01 mark

$Alternatively$

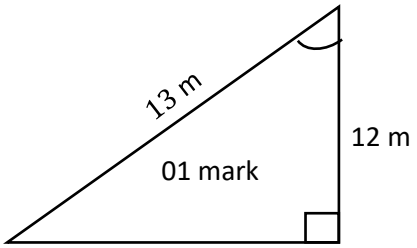
$\frac{AC}{AN} = \frac{AB}{AM} = \frac{BC}{MN} = \frac{3}{2} = Constant\ of\ proportionality$ ……01 mark

$< BAC \cong < MAN\ (Common\ angle\ )$……01 mark

$\therefore \Delta\ ABC \sim \Delta\ AMN(AA - Similarity\ theorem)$ ……01 mark

9. (a)

Figure 1



By Pythagoras theorem$x$

$x^2 = 13^2 - 12^2$.....01 mark

$x^2 = 169 - 144$

$x^2 = 25$.....01 mark

$\sqrt{x^2} = \sqrt{25}$

$x = \pm 5$ .....01 mark

$x = 5, x \geq 0$

Therefore the foot of the ladder is 5 m away from the wall .....01 mark

(b) $\cos 60^\circ = \frac{9}{x}$ .....01 mark

$x = \frac{9}{\cos 60^\circ}$

$x = \frac{9}{.5}$.....01 mark

$= \frac{9\times 10}{.5\times 10}$.....01 mark.

$= \frac{90}{5}$.....01 mark.

$= 18\ cm$.....01 mark

10. (a) i. Total tracksuits sold $= 20 + 10 + 80 + 120 + 180$

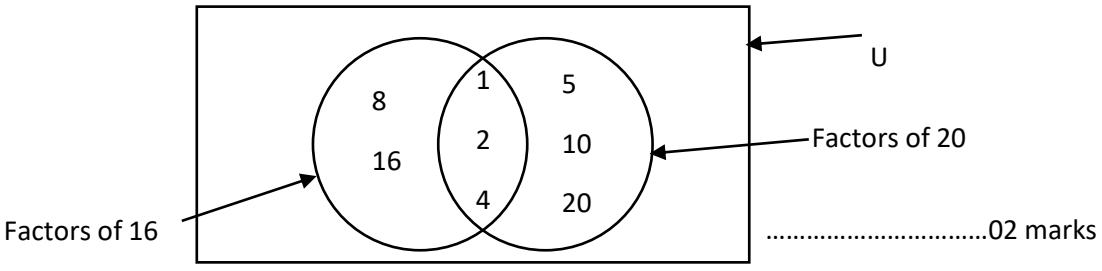
$= 410$ tracksuits .....01 mark

ii. $180 - 10 = 170$

Therefore 170 more tracksuits were sold in November than in August.....01 mark

(b) i. Factors of 16 are 1, 2, 4, 8, and 16

Factors of 20 are 1, 2, 4, 5, 10, and 20



.....02 marks

ii. The highest common factor of 16 and 20 is 4.....01 mark

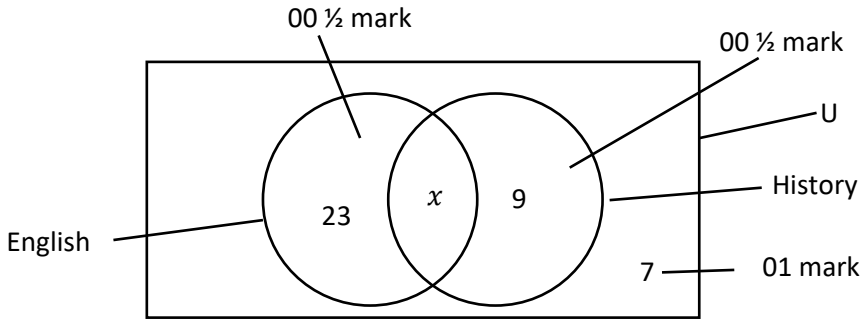
(c) Let the number of students in the class be μ

Let the number of students who study English be E

Let the number of students who study History be H

$n(\mu) = 44, n(E \cap H') = 23, n\,(E' \cap H) = 9, n(E \cap F)' = 7$

Figure 3



$23 + x + 9 + 7 = 44$

$x = 44 - 39$

$x = 5$……01 mark

$n(E) = 23 + 5$

$= 28$

Therefore 28 students study history……01 mark.